

TRELLIS.2 Printability Fine-Tuning - Weekly Report

1. Executive summary

We retired the inference-time DPO steering approach, established that supervised fine-tuning is the better instrument for this problem, built the full data and training pipeline, and completed a 1,170-step LoRA fine-tune of TRELLIS.2's shape-SLat flow model on 300 clean Thingi10K models.

The fine-tune works on the axes it can reach. Measured on exported meshes, held-out set, paired sampling noise:

Model	L_OH (overhang)	L_Th (thin walls)	L_Topo	R_Detail
Base model	0.2291	0.1133	0.1871	0.4316
step 750	0.2183 (-5%)	0.0152 (-87%)	0.0500 (-73%)	0.4578 (+6%)
step 900	0.2255 (-2%)	0.0299 (-74%)	0.0635 (-66%)	0.4411 (+2%)
step 1050	0.2179 (-5%)	0.0159 (-86%)	0.0340 (-82%)	0.4358 (+1%)

Step 1050 beats the base model on thin-wall penalty for 6 of 6 held-out objects. Image fidelity did not degrade (0.897 -> 0.908-0.913), and surface detail slightly increased, so this is not printability bought by smoothing the model into generic blobs.

2. What was discovered

2.1 The retired approach failed for a diagnosable reason

Inference-t
17 scored
(slat_detail
a 33% cha
near-ortho

2.2 The "differentiability wall" did not exist

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only conne
inspection

2.3 The VAE bounds what any flow-model training can achieve

Encoding a

Property	Result
Shape fidelity	Preserved - Chamfer 0.0052, p95 0.0101, bounding box exact
Watertightness	0 of 2 preserved
Defect rate (full mesh)	0.086% open edges, 0.095% non-manifold
Components	953, but largest is 99.85% of faces - and still not watertight alone

Consequence: geometry-carried printability (thickness, overhang, bulk shape) is reachable by training the flow model; watertightness is not, and belongs to post-processing. This single 20-minute measurement redirected the entire project.

2.4 The judge inflates topology defects ~50x

geometric_
rate on the
L_Topo fig
defects. Ar

2.5 Measurement noise nearly made the whole run uninterpretable

Two evalua
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held-out ob
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2.6 Silent failures dominated the engineering cost

Every sign

Symptom	Actual cause	Cost
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Training step took 464s	No activation checkpointing -> 38 GB retained	18x slowdown
Baseline eval returned n_eval=0	Three stacked bugs, masked by except: contin	2 invalidated launches
Eval decoder on CPU	low_vram is device placement, not a memory hi	1 launch
Sample-evolution panel empty	A patch silently failed to match; the call site was	whole run, cosmetic
Loss chart nonsense	Dataset build logged token counts into the loss f	misleading chart

The single highest-leverage change all week was replacing except Exception: continue with recorded, printed failures. It converted two blind retries into one-step diagnoses.

3. What was built

Component	File	Note
CPU mesh -> SLat encoder	backends/dual_grid_cpu.py	o-voxel ships CUDA-only; rebuilt the pure-C++
Offscreen renderer	trellis_core/render_mesh.py	moderngl, Metal-backed, headless; camera fitt
Dataset pipeline	trellis_core/dataset_build.py	300 models, 3.2 GPU-hours, 0 failures
LoRA	trellis_core/lora.py	210 modules, 18.68M params (1.4% of 1.29B)
SFT trainer	trellis_core/sft_train.py	Hyperparameters from TRELLIS.2's own config
Live dashboard	devlab/train_server.py	Crash-independent; phone access over Tailsca
VAE round-trip test	trellis_core/vae_roundtrip.py	Established the ceiling
Mesh exporter	trellis_core/export_samples.py	Reproduces eval exactly (same seeds, same
Topology test	trellis_core/topology_test.py	Paired, undecimated, multi-seed

Dataset

Thingi10K
3D-printing
multi-comp
PWN + no
(270 train /

Training run

steps	1,170 of 1,200 (stopped after divergence)
epochs	8.7 over 270 examples
GPU time	9.96 hours, mean 30.7 s/step
LoRA	rank 16, alpha 32, lr 1e-4 constant, AdamW wd 0.01
schedule	uniform t, p_uncond 0.1 (both from TRELLIS.2's official config)
evals	10, every 150 steps, 6 held-out objects, common random numbers

run id

4. The divergence

Steps	Mean loss	Max loss	Grad norm
1-1100	~0.16	0.80	0.03-0.06
1101-1170	3.48	57.29	332.25

Held-out loss went 0.0859 -> 10.29; image similarity 0.908 -> 0.202. The model was destroyed.

Two causes, both mine: no learning-rate decay (constant 1e-4 for 8.7 epochs), and gradient clipping that bounded each update but could not prevent AdamW's moments accumulating through a sustained blow-up. Checkpoints 750/900/1050 were preserved before stopping, because a clean stop writes a final checkpoint and prune_checkpoints(keep=3) would have deleted step 750.

5. How to log improvements numerically

Recommended standard for every future comparison. The first three are non-negotiable given what we measured this week.

5.1 Always paired, never pooled

Fix the sam
per-object
is between
within-fork

5.2 Always against a baseline

Every run
number wi

5.3 Always report fidelity beside printability

Minimum p
conditionin
experimen

5.4 The metric set

				Metric
L_Th	judge	lower	Responds strongly. Primary succe	
nonmanifold_rate, open_rate	full mesh	lower	Must be computed undecimated (	
L_OH	judge	lower	Has not responded; also only ~5%	
R_Detail	judge	context	Fidelity term, not printability - no be	
image_similarity	silhouette IoU	higher	Prompt-adherence guard	
sample_dispersion	latent means	flat	Collapse guard	
watertight_rate	full mesh	control	Expected flat; a change indicates a	
support volume	slicer, not yet built	lower	The only metric outside the trainin	

5.5 The acceptance bar

Improveme

```
capture = (tuned_1_sample - base_1_sample) / (base_best_of_N - base_1_sample)
```

A single sample from the tuned model must recover a meaningful fraction of what N-fold rejection sampling buys from the base model at matched compute. This test has not yet been run.

6. Next steps

1. Finish the topology consistency test (running: 10 objects x 3 seeds x 4 models, paired, undecimated). Answers whether step 1050's cleaner topology is consistent or a lucky draw.
2. Run the best-of-N comparison (§5.5). Until this exists we do not know whether the fine-tune beats cheap rejection sampling - the only comparison that justifies the work.
3. Fix the training schedule and re-run. Cosine LR decay; a skip-step rule that drops a batch whose gradient norm exceeds a multiple of the running median rather than clipping it; keep every checkpoint. The metrics were still improving when the run broke, so the ceiling was never observed.
4. Repair the judge before it is trusted further: make the thickness raycast deterministic (test-retest reliability was measured at 0.208, capping any correlation at $|r| \leq 0.456$), and add a support raycast to the overhang term (currently rank-correlates -0.432 with true unsupported area).
5. Scale the dataset. 270 training objects is the binding constraint, not step count; 3,935 clean models are available and prep runs at ~40 s/model.
6. Add a slicer-based support-volume metric as the held-out physical measure that was never part of the training reward.

7. Deliverables on disk

```
runs/sft-1200-20260806-0119/
  ckpt_keep/step_{750,900,1050}/adapter.pt    preserved LoRA adapters (643 MB)
  meshes/step_{0,750,900,1050}/               24 GLB + 24 STL, raw, no post-processing
  exports/                                     42 PNG frames + 6 progression strips
```

metrics.jsonl	1,170 train + 10 eval records
topology_test.jsonl	in progress
data/thingi10k_sft/	300 latents + images + conditioning

All meshes are raw decoder output - no repair, no decimation, no solidify. Each object uses identical conditioning and identical sampling noise across all four models, so differences are attributable to the model alone.